

Erratum to “Quantifying Competitive Relationships Among Open-Source Software Projects”

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Paper Information

Title: Quantifying Competitive Relationships Among Open-Source Software Projects

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Description of the Error

After submission, we identified an implementation error in the script used to construct the decision-tree features for RQ1 and RQ2.

The MIAO score from project i to project j was intended to be calculated by aggregating the corresponding SCE values across all analysis periods:

$$MS_{ij} = - \sum_{k=1}^m SCE_{ji}^k. \quad (1)$$

However, the original feature-generation script incorrectly used only the SCE value from the final analysis period:

$$MS_{ij} = -SCE_{ji}^m. \quad (2)$$

Consequently, the decision-tree classifiers were originally trained and evaluated using features that did not correspond to the intended MIAO-score aggregation.

Correction

We corrected the feature-generation script so that each MIAO-score feature is calculated by aggregating the SCE values across all applicable analysis periods.

For EVAL1, the corrected features aggregate the SCE values over the full observation period. For EVAL2, which evaluates one-year-ahead prediction, the corrected features aggregate only the analysis periods available before the prediction point.

We then recalculated all decision trees used in Sections 5.2 and 5.3 of the paper.

Corrected Results

The correction affects the decision-tree-based results and their interpretation. It does not affect the proposed MIAO framework, the dataset construction, the research questions, or the evaluation design.

For EVAL1, the corrected maximum accuracy is 78%, and the corrected mean accuracy is 73%. The mean REV F1 score remains 71%, unchanged from the originally reported result.

For EVAL2, the corrected maximum one-year-ahead accuracy is 74%, and the corrected mean accuracy is 68%.

Table 1: Comparison of the originally reported and corrected results.

Evaluation	Maximum accuracy		Mean accuracy	
	Original	Corrected	Original	Corrected
EVAL1	81%	78%	75%	73%
EVAL2	77%	74%	73%	68%

Correction to the RQ1 Findings

The originally reported accuracy values for both EVAL1 and EVAL2 should be replaced with the corrected values shown above.

Although the corrected accuracy values are lower, the corrected analysis continues to support the conclusion presented for RQ1. EVAL1 retains a mean REV F1 score of 71%, and EVAL2 reaches a maximum one-year-ahead accuracy of 74%.

The corrected results therefore continue to indicate that features derived from MIAO provide useful signals for distinguishing REV cases. However, the predictive evidence, particularly the mean performance of EVAL2, is weaker than originally reported.

Correction to the RQ2 Findings

In the original analysis, the primary decision-tree patterns were interpreted as follows:

- EVAL1 primarily identified unidirectional influence from the target project to its competitors.
- EVAL2 primarily identified unidirectional influence from the competitors to the target project.

The corrected analysis produces the following findings:

- For EVAL1, unidirectional influence from the target project to its competitors remains the primary signal. This finding is unchanged from the original analysis.
- For EVAL2, unidirectional influence from the target project to its competitors becomes the primary signal, with additional contributions from influence in other directions.

Therefore, the substantive correction for RQ2 concerns EVAL2. Its primary signal changes from competitor-to-target influence to target-to-competitor influence, while its decision trees also contain a more heterogeneous combination of directional relationships.

Impact on the Conclusions

The correction reduces the reported classification performance and changes part of the interpretation of the key competitive-influence directions.

For RQ1, the corrected results preserve the paper’s overall conclusion that MIAO-derived features provide useful signals for identifying REV cases. Nevertheless, the corrected values must be used in place of the originally reported accuracy values, and the lower mean accuracy for EVAL2 should be taken into account when assessing the strength of the predictive results.

For RQ2, the EVAL1 conclusion remains unchanged. The interpretation of EVAL2 must be revised because its primary signal changes from competitor-to-target influence to target-to-competitor influence, with some contribution from other directions.

Despite this change, the corrected results continue to indicate that MIAO captures differences and temporal changes in the structure of competitive relationships among OSS projects. The correction does not change the definition of MIAO, the underlying SCE calculations, the dataset, or the evaluation procedure, but it does require revision of the reported decision-tree performance and the EVAL2 feature-level interpretation.

Corrected Artifact

The corrected artifact, including the corrected implementation and recalculated experimental results, is archived in the following Zenodo record:

<https://zenodo.org/records/21766491>

For convenience, we also provide a directly executable demonstration notebook in the project repository:

<https://github.com/noppoMan/MIAO/blob/main/demo.ipynb>

The notebook provides a simpler way to reproduce and inspect the corrected MIAO-score calculation and the resulting decision-tree analysis. The Zenodo record should be regarded as the archival reference for the corrected artifact.

Corrected Manuscript

The corrected version of the manuscript is available on arXiv:

<https://arxiv.org/abs/2602.17131>

Acknowledgment

We sincerely apologize for this implementation error and for any confusion it may have caused. We provide this correction proactively to ensure that the published record, manuscript, and associated artifacts accurately reflect the intended MIAO-score calculation and the corrected experimental results.